

SESSION 5

Responsive Web Design

Responsive Principles · Media Queries · Mobile-First · Components · Layout Techniques

Design Principles

Media Queries

Mobile-First

Components

Layout Techniques

Prepared & Presented by

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Technical Trainer | Frontend Essentials Bootcamp 2026

About the Presenter



Utkarsh Kushwaha

Full Stack Developer | Technical Speaker | Community Mentor

Experienced in building responsive and interactive web applications using modern technologies and actively involved in technical communities, workshops, and student learning initiatives.

TECHNICAL SKILLS

HTML

CSS

JavaScript

React

Node.js

MongoDB

Express

Docker

Connect:



LinkedIn



GitHub

Learning Objectives

By the End of This Session, Students Will Be Able To:

- ✓ Understand core responsive design principles — fluid grids, flexible images, relative units
- ✓ Apply the mobile-first development approach when writing CSS
- ✓ Use Flexbox and Grid together for modern responsive layouts
- ✓ Write media queries to adapt layouts across breakpoints
- ✓ Build responsive components like navbars and card grids
- ✓ Structure a fully responsive webpage layout end-to-end

Outcome: Students will be able to design and build webpages that adapt effectively across screen sizes and devices.

Session Agenda



Responsive Design Principles

fluid grids, flexible units



Media Queries

breakpoints & syntax



Mobile-First Approach

build small screens first



Responsive Components

navbar & card grid



Layout Techniques

flexbox + grid combined



Mini Project

responsive landing page

Quick Revision

Previously Learned in Session 4

- Display Properties (block, inline, inline-block, none)
- Flexbox — Container & Item Properties
- Navigation Bar Design
- Card Components
- Section-Based Layouts
- Footer Design

CODE

```
.navbar {  
  display: flex;  
  justify-content: space-between;  
  align-items: center;  
}
```

Outcome: Today we take those fixed layouts and make them adapt to every screen size.

01

Responsive Design Principles

Building Webpages That Adapt to Any Screen

What Makes a Design “Responsive”?

A responsive layout adjusts itself using flexible units — not just breakpoints.



Fluid Grids

Use % or fr units instead of fixed px widths



Flexible Images

max-width: 100% so images scale with their container



Relative Units

rem, em, % and vw/vh instead of fixed px values



Viewport Meta Tag

Tells mobile browsers to render at device width

CODE

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

Outcome: Responsive design starts with flexible units and the viewport tag — not just media queries.

02

Media Queries

Applying Different Styles for Different Screen Sizes

Media Query Syntax & Breakpoints

Applies CSS rules only when a condition about the screen is true.

`@media syntax`

Wraps a block of CSS rules inside a condition

`max-width / min-width`

Apply styles below or above a given screen width

`Common breakpoints`

480px (mobile) · 768px (tablet) · 1024px (desktop)

`Query order`

Mobile-first uses min-width; desktop-first uses max-width

CODE

```
@media (max-width: 768px) {  
  .container { flex-direction: column; }  
}
```

Outcome: Media queries let one stylesheet serve phones, tablets, and desktops.

03

Mobile-First Development

Design for Small Screens First, Then Scale Up

Mobile-First vs Desktop-First

Two ways to order your CSS — pick one and stay consistent.



Mobile-First

- Base CSS = mobile styles
- Add complexity with min-width queries
- Leaner CSS, faster on phones



Desktop-First

- Base CSS = desktop styles
- Simplify with max-width queries
- Common in older codebases

CODE — MOBILE-FIRST

```
.card { width: 100%; }  
@media (min-width: 768px) {  
  .card { width: 50%; }  
}
```

Outcome: Start with mobile styles as the default, then add complexity with min-width queries.

04

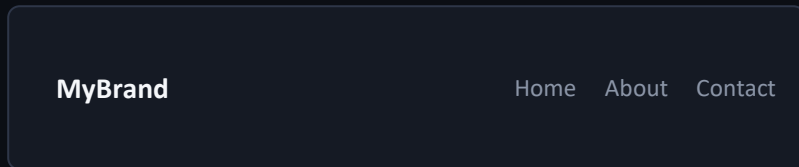
Responsive Components

Turning Static Layouts Into Adaptive Ones

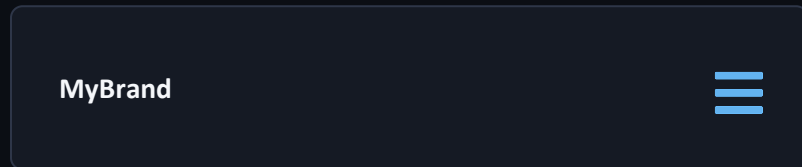
Building a Responsive Navigation Bar

Hide nav links behind a hamburger icon on small screens.

LIVE PREVIEW



Desktop — links visible



Mobile — links collapse into icon

CODE

```
@media (max-width: 768px) {  
  .nav-links { display: none; }  
  .hamburger { display: block; }  
}
```

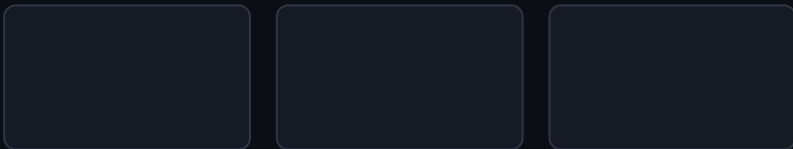
Outcome: Toggle `.nav-links` off and `.hamburger` on inside the same media query.

Building a Responsive Card Grid

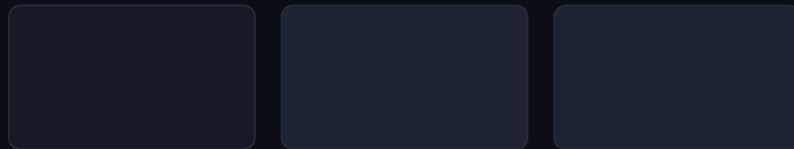
Grid columns collapse automatically once you adjust them inside a media query.

LIVE PREVIEW

Desktop — 3 columns



Mobile — 1 column



CODE

stacks vertically below 768px

```
.grid {  
  display: grid;  
  grid-template-columns: repeat(3, 1fr);  
}  
  
@media (max-width: 768px) { .grid { grid-template-columns: 1fr; } }
```

Outcome: Redefine grid-template-columns inside a media query to reflow the whole grid.

05

Responsive Layout Techniques

Flexbox + Grid Working Together

CSS Grid: auto-fit & minmax()

Builds a responsive grid without writing a single media query.

`repeat(auto-fit, ...)`

Fits as many columns as the available space allows

`minmax(220px, 1fr)`

Each column is at least 220px, then grows to fill space

No breakpoints needed

The grid reflows automatically as the screen shrinks

CODE

```
.grid {  
  display: grid;  
  grid-template-columns: repeat(auto-fit, minmax(220px, 1fr));  
}
```

Outcome: auto-fit + minmax() is the fastest way to build a responsive grid without extra breakpoints.

Best Practices in Responsive Design



Mobile-first by default

Write base styles for small screens, then scale up



Use relative units

Prefer rem, em and % over fixed px values



Test real breakpoints

Check at 480px, 768px, 1024px and 1440px



Avoid fixed widths

Prefer max-width paired with width: 100%



Combine Flexbox & Grid

Flex for rows and navbars, Grid for card layouts



Always add the viewport tag

Without it, mobile responsiveness silently fails

Mini Project: Responsive Landing Page

Combine everything from today into one adaptive page.

`navbar`

`hero section`

`feature cards × 3`

`footer`

Build this layout using:

- A proper viewport meta tag
- Media queries at the 768px breakpoint
- Mobile-first CSS as the base styles
- A CSS grid using auto-fit + minmax()

Concepts used: media queries, mobile-first, grid, flexbox

Today We Learned



Responsive Principles

fluid grids, flexible images



Media Queries

breakpoints & syntax



Mobile-First

build small screens first



Responsive Navbar

hamburger menu pattern



Responsive Grid

auto-fit + minmax()



Layout Techniques

flexbox + grid combined

Outcome: You can now build webpages that adapt smoothly across phones, tablets, and desktops.

Thank You!

Questions? I'm happy to help!

Practice Task:

Make your Session 4 landing page fully responsive — add a viewport meta tag, a mobile hamburger navbar, and a CSS grid that collapses to one column below 768px.

Session 6 — next Saturday on Discord · we build on this with real project layouts